1 **(i)** Without using a calculator, solve the inequality

$$\frac{x+2}{x-1} \le \frac{x}{x+3}.$$ [4]

(ii) Hence find the set of values of x for which $\dfrac{e^x+2}{e^x-1} \le \dfrac{e^x}{e^x+3}$. [2]

2 A curve C has parametric equations

$$x = t - \cos 2t \quad y = \sin^2 t,$$

where $0 \le t \le \dfrac{\pi}{3}$.

Show that the area under C for $0 \le t \le \dfrac{\pi}{3}$ is $\displaystyle\int_a^b \left(\sin^2 t + 2\sin^2 t \sin 2t\right) \mathrm{d}t$, where a and b are constants to be determined. Hence find the exact value of this area. [6]

3 **(a)** Given that $\displaystyle\int_0^b \frac{3}{\sqrt{1-4x^2}}\,\mathrm{d}x = \frac{\pi}{2}$, find the exact value of b. [3]

(b) Using the substitution $x = 2\tan\theta$, where $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$, find $\displaystyle\int \frac{1}{\sqrt{4+x^2}}\,\mathrm{d}x$. [4]

4 Relative to the origin O, the points A and B have position vectors **a** and **b** respectively, and **b** = k**a**, where k is a positive real constant. The circle, with centre C and AB as its diameter, has O lying outside it. T is a point on this circle with position vector **t** and OT is a tangent to the circle. (See Diagram 1)

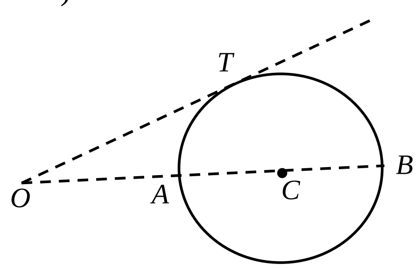

Diagram 1

(i) By considering the angle ATB, show that $\mathbf{a}\bullet\mathbf{t} = \dfrac{k|\mathbf{a}|^2 + |\mathbf{t}|^2}{k+1}$. [3]

(ii) By considering the angle OTC and using the result in part **(i)**, determine the ratio of $|\mathbf{a}|:|\mathbf{t}|$, giving your answer in terms of k. [4]

5 Matchsticks are used to form a pattern of n rows of triangles in n stages. In the 1st stage, one triangle is formed using 3 matchsticks. In the 2nd stage, three more triangles are added to the second row with an addition of 6 matchsticks. In the 3rd stage, five more triangles are added to the third row with an addition of 9 matchsticks, as shown in the Diagram 2 below. The process continues till the n^{th} stage is completed with n rows of triangles formed.

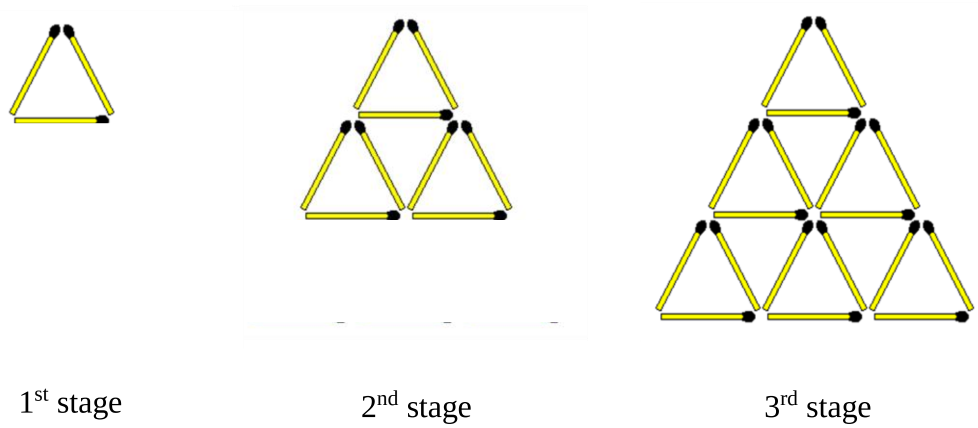

Diagram 2

Let u_r be the total number of matchsticks used after the r^{th} stage is completed, with a pattern of r rows of triangles formed.

(i) Write down u_1, u_2 and u_3. [1]

(ii) The relation between u_{r+1} and u_r is given by $u_{r+1} = u_r + ar + b$, where $r \in \mathbb{Z}^+$. Use the information in part **(i)**, show that $a = b = 3$. [2]

(iii) Find $\sum_{r=1}^{n-1}(3r+3)$ in terms of n. Hence using the relationship given in part **(ii)**, find u_n in terms of n. [5]

6 The function f is defined by

$$\mathrm{f}: x \mapsto \left|\ln(x+3)\right|, \quad x \in \mathbb{R}, -3 < x \le k, \text{ where } k \text{ is a real constant.}$$

(i) Explain why f^{-1} does not exist when $k = 0$. [1]

(ii) State the maximum value of k such that f^{-1} exist. [1]

Let k be the value found in part **(ii)** for the rest of this question.

(iii) Find $\mathrm{f}^{-1}(x)$ in similar form. [3]

(iv) The function g is defined by

$$\mathrm{g}: x \mapsto a + \mathrm{e}^{x}, \quad x \in [-\ln 2, \infty), \text{ where } a \text{ is a constant.}$$

Show that the composite function gf exists and determine the range of gf in terms of a. [3]

7 The curve C_1 has equation $y = \mathrm{e}^{x}$, where $x \ge 0$. The curve C_2 has equation $y = 1 + \mathrm{e} - x^2$, where $x \ge 0$. The region bounded by C_1, C_2 and the y-axis is denoted as R.

(i) Using integration, show that $\int (\ln x)^2 \, \mathrm{d}x = x(\ln x)^2 - 2x\ln x + 2x + C$, where C is an arbitrary constant. [3]

(ii) Verify that the point of intersection of C_1 and C_2 is (1, e). [1]

(iii) Hence, or otherwise, find the exact value of the volume of revolution formed when R is rotated through 4 right angles about the y-axis. [5]

8 The line l_1 has equation $\mathbf{r} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, where $\lambda \in \mathbb{R}$. The plane Π has equation $x + 2y - z = 4$.

(i) Find the acute angle between l_1 and Π. [2]

(ii) Find the position vector of the point where l_1 intersects Π. [2]

(iii) Find the foot of perpendicular from point $A(2,2,3)$ to Π. Hence find a cartesian equation of the line which is a reflection of l_1 in Π. [5]

The line l_2 has an equation $\mathbf{r} = (-1-3\mu)\mathbf{i} + (4+\mu)\mathbf{j} + (7+\mu)\mathbf{k}$, where $\mu \in \mathbb{R}$.

(iv) Determine if the lines l_1 and l_2 are parallel, intersecting or skew lines. [2]

9 A curve C has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = \frac{x-1}{a} + \frac{a}{x-1}, \quad x \in \mathbb{R},\ x \neq 1,$$

with a being a constant such that $-1 < a < 0$.

(i) Find algebraically, the set of values that $\mathrm{f}(x)$ can take. [3]

(ii) Sketch C, showing clearly the equations of any asymptotes and coordinates of any axial intercepts and stationary points in terms of a. [4]

(iii) On separate diagrams, sketch the graph of

(a) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(b) $y = \mathrm{f}'(x)$, [2]

showing clearly in each case, the equations of any asymptotes and the coordinates of any axial intercepts and stationary points in terms of a.

(iv) Describe a sequence of transformations that transform the graph of $y = \mathrm{f}(x)$ to $y = \dfrac{x}{2a} + \dfrac{a}{2x}$. [2]

10 A mine-sweeping robot is used to sweep mines in a mine-field. The robot is programmed to move through the mine-field in a particular manner. From its starting position denoted as O in the mine field, the robot will move in a straight line covering a distance of 100 metres due east, and then it will turn through an angle of 90°in an anti-clockwise direction. After making the first turn, it will travel in a straight line covering a distance of 80 metres, and then turn through an angle of 90° in an anticlockwise direction. The robot will repeat this process of moving in a straight line and turn through an angle of 90° in an anti-clockwise direction throughout its motion, and the distance covered by the robot between the n^{th} turn and the $(n+1)^{\text{th}}$ turn is 20% less than the distance covered by the robot between the $(n-1)^{\text{th}}$ turn and the n^{th} turn, with $n \in \mathbb{Z}^+$, as shown in Diagram 3 below.

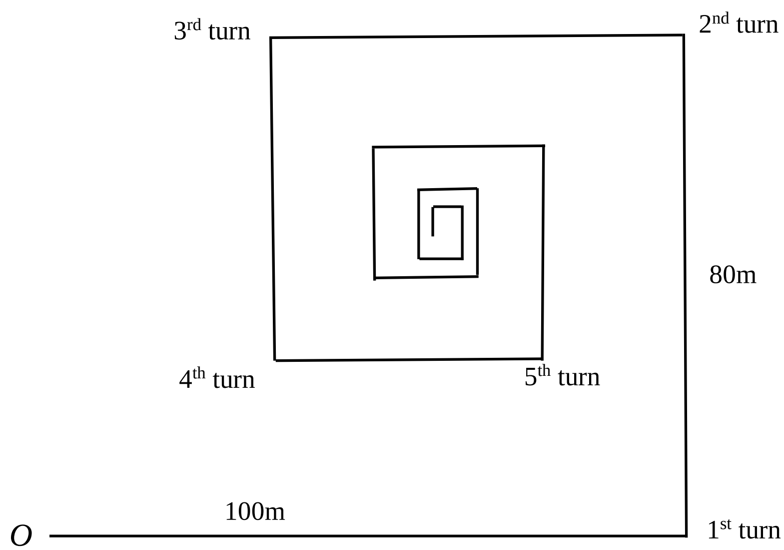

Diagram 3

(i) Show that the distance covered by the robot between its 11^{th} turn and 12^{th} turn is 8.59 metres, correct to 3 significance figures. [2]

(ii) Find the number of turns the robot has made after covering a total distance of 485 meters. [3]

(iii) Determine the coordinates of the theoretical final position that the robot will end up with respect to O. [4]

After a change in the robot's setting, the distance covered by the robot between the n^{th} turn and the $(n+1)^{\text{th}}$ turn is x % less than the distance covered by the robot between the $(n-1)^{\text{th}}$ turn and the n^{th} turn, $n \in \mathbb{Z}^+$. Given that the initial distance covered by the robot remains as 100 metres, and the robot will cover a total distance of 500 metres just before making its 9^{th} turn, determine the value of x. [3]

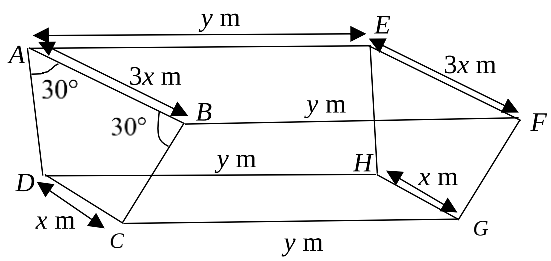

Diagram 4

A farmer requires an open trough to be made of wood of negligible thickness (see Diagram 4). The trough is to contain $\dfrac{20}{\sqrt{3}}$ m^3 of liquid feed when it is full. The cross-section of the trough *ABCD* is an isosceles trapezium with *AB* being $3x$ metres long and *DC* being x metres long, and $\measuredangle DAB = \measuredangle ABC = 30°$. The cross-section *ABCD* is perpendicular to both the rectangular base *DHGC* and the rectangular opening *AEFB*, with $AE = BF = DH = CG = y$ metres.

(i) Show that the external surface area A of the trough is $A = \dfrac{4x^2}{\sqrt{3}} + \dfrac{40\sqrt{3}+30}{3x}$. [4]

(ii) Using differentiation, find the value of x when A is minimum. [4]

(iii) Due to a crack at the base of the trough, liquid feed contained in the trough leaked out at a constant rate of $0.01\ \text{m}^3$ per hour. Given that $y = 5$ and the trough is initially full of liquid feed, find the rate at which the depth of the liquid feed is decreasing after 72 hours. [4]

End of Paper